

Enterprise AI Architecture and Governance

Introduction

Enterprise AI systems combine large language models, vector databases, orchestration frameworks, observability layers, and governance controls. Organizations deploying AI systems must establish strict security boundaries, prompt filtering, role-based access controls, and monitoring pipelines to prevent data leakage and ensure compliance.

RAG Pipeline

Retrieval-Augmented Generation pipelines typically involve document ingestion, chunking, embedding generation, vector indexing, semantic retrieval, reranking, and final answer synthesis. Frameworks such as LangChain and LlamaIndex are often integrated with embedding models and vector stores like FAISS, Pinecone, or ChromaDB.

Challenges

Common challenges include hallucination reduction, token optimization, embedding drift, metadata consistency, latency bottlenecks, and scaling retrieval across distributed systems. Engineers frequently optimize chunk overlap strategies and embedding dimensions to improve retrieval accuracy.

Security

Security controls include encryption at rest, secure API gateways, IAM role segregation, audit logging, content moderation, and zero-trust network architecture. Enterprises also implement rate limiting and anomaly detection to mitigate misuse of generative AI systems.

Component	Purpose	Technology Example
Embedding Model	Vector Representation	SentenceTransformers
Vector Database	Semantic Retrieval	ChromaDB / FAISS
LLM	Answer Generation	Claude / GPT / Titan
Orchestration	Workflow Management	LangChain